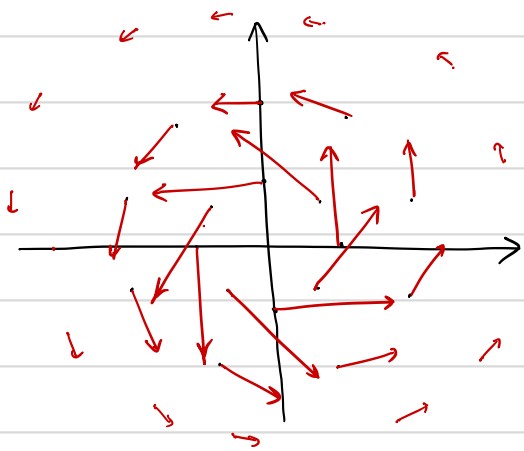
This is only possible when $\vec{F}=\langle P,Q\rangle$ is undefined at certain points of the plane.

Supplementary notes

Example where $Q_x = P_y$, but $\langle P,Q\rangle \neq \nabla f$ for any $f=f(x,y)$.

$$\vec{F} = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}\right) = \frac{\langle -y, x\rangle}{|\langle -y,x\rangle|^2}$$



$$\frac{\partial Q}{\partial x} = \frac{1\cdot(x^2+y^2) - x\cdot 2x}{(x^2+y^2)^2}$$

$$= \frac{-x^2+y^2}{(x^2+y^2)^2}$$

$$\frac{\partial P}{\partial y} = \frac{-1\cdot(x^2+y^2) - (-y)\cdot 2y}{(x^2+y^2)^2}$$

$$= \frac{-x^2+y^2}{(x^2+y^2)}$$

So: $\dfrac{\partial Q}{\partial x} = \dfrac{\partial P}{\partial y}$ $\quad (\operatorname{curl}(\vec{F}) = 0)$

But: $\vec{F}$ is not conservative!

(not a gradient)

Let's check:

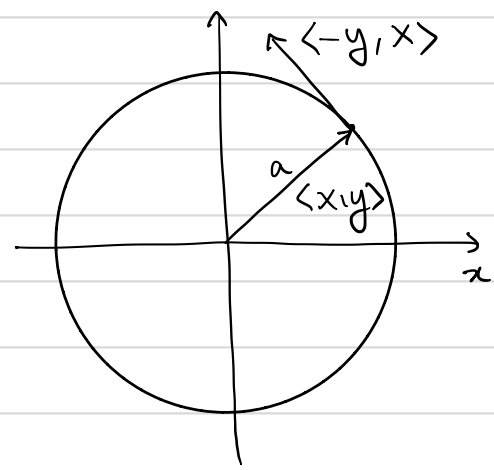
$$\int_C \vec{F}\cdot d\vec{r} = 2\pi \ (\neq 0),$$

$\Downarrow$

$\vec{F} \neq \nabla f$

where

C = circle of radius a with center $(0,0)$.



Parametrize C:

$$\begin{cases} x = a\cos t & dx = -a\sin t\, dt \\ y = a\sin t & dy = a\cos t\, dt \end{cases}$$

$(0 \leq t \leq 2\pi)$

$$\vec{F} = \left\langle \frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}\right\rangle$$

$$= \left\langle \frac{-a\sin t}{a^2}, \frac{a\cos t}{a^2}\right\rangle$$

$$= \frac{1}{a}\langle -\sin t, \cos t\rangle = \langle P, Q\rangle$$

$$\int_C \vec{F}\cdot d\vec{r} = \int_C \langle P, Q\rangle \cdot \langle dx, dy\rangle$$

$$= \int_0^{2\pi} \frac{1}{\cancel{a}} \langle -\sin t, \cos t\rangle \cdot \langle -\cancel{a}\sin t, \cancel{a}\cos t\rangle dt$$

$$= \int_0^{2\pi} \underbrace{(-\sin t)^2 + (\cos t)^2}_{1} \, dt = 2\pi //$$